Travis Beckwith, Ph.D.

Bloomington, IN 47408 | Phone: +1 330-321-2517 | Email: travis.beckwith@gmail.com
ORCID: https://orcid.org/0000-0001-6128-8464 | LinkedIn: https://www.linkedin.com/in/travis-beckwith-phd
GitHub: https://github.com/TravisBeckwith | Website: https://travisbeckwith.github.io/CV
Google Scholar: https://scholar.google.com/citations?user=wolY848AAAAJ&hl=en (Citations: 399, h-index: 10)

PROFESSIONAL SUMMARY

Neuroscientist who studies how childhood environmental exposures and experiences shape brain development and decision-making. Training in human and rodent research methodologies. Principal imaging analyst for three NIH-funded longitudinal cohorts, applying mixed-effects models to structural, functional and diffusion imaging alongside behavioral and epidemiological measures. Has taught graduate and high school students, mentored first-year undergraduates (Springboard Program, BGSU), and holds a Preparing Future Faculty certificate. Builds open-source analysis tools and has evaluated frontier AI systems for accuracy in neuroscience, biology and clinical medicine. Author of 11 peer-reviewed publications (h-index 10, 399 citations).

RESEARCH & PROFESSIONAL INTERESTS

- **Early Environment and Decision-Making:** How early-life environment, including caregiving and enrichment, shapes brain development and decision-making, using rodent models and human developmental and neuroimaging data.
- **Brain-Behavior Relationships**: Neural mechanisms underlying cognitive and behavioral outcomes across environmental health, neurotrauma, and clinical populations.
- **Quantitative Methods and Open Science:** Reproducible multimodal MRI analysis of structural, functional, and diffusion data, open-source tools, and machine learning, with attention to evaluating evidence, including AI-generated output.

EDUCATION

Ph.D., Neuroscience | 2008–2015
University of Cincinnati College of Medicine, Cincinnati, OH

- Dissertation: "A Magnetic Resonance Imaging Study of the Developmental Consequences of Childhood Lead Exposure in Adulthood"
- Committee: Kim Cecil, PhD (Advisor); Kim Dietrich, PhD; James Eliassen, PhD; James Herman, PhD; Michael Williams, PhD

B.A., Psychology & Neuroscience; B.F.A., Creative Writing & English | 2001–2007
Bowling Green State University, Bowling Green, OH

- Graduated cum laude with University and English Honors

Additional Training

Post-Baccalaureate Certificates

- **Clinical and Translational Research** | 2018–2019
  University of Cincinnati College of Medicine, Cincinnati, OH

- **Preparing Future Faculty Program** | 2012–2013
  University of Cincinnati, Cincinnati, OH

Neuroimaging Methods Training

- CONN Toolbox Functional Connectivity Workshop | 2015

Workshops

- NIH Grant Writing Workshops | May 2020 | Cincinnati Children's Hospital Medical Center
  - Planning and Writing Successful NIH Career Development (K) Awards
  - Planning and Writing Successful NIH R01 Renewals and Submissions

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TECHNICAL SKILLS

Neuroimaging Platforms: FSL • MRtrix3 • DESIGNER2 • AFNI • SPM (including CAT12) • FreeSurfer • ANTs • fMRIPrep • QSIPrep • QSIRecon • AMICO • CONN • ImageJ • dcm2niix • E-Prime • PsychoPy

Modalities: fMRI (task-based & resting-state) • Diffusion MRI (DTI, CSD, multi-shell) • NODDI microstructure modeling • MRS • Structural/Volumetric • Surface-based morphometry • Structural connectomics • FDG-PET/CT (limited experience)

Analysis Methods: CSD tractography (multi-tissue, single- and multi-shell) • SIFT2 streamline filtering • NODDI microstructure estimation (NDI, ODI, ISOVF) • Atlas-based connectome construction (Desikan-Killiany) • Multi-weighted connectivity matrices (streamline count, FA, length) • Cortical thickness analysis • Cortical surface area & gyrification • Volumetric segmentation • White matter tractography (deterministic & probabilistic) • Resting-state connectivity (ICA) • Brain-behavior correlation • Longitudinal progression tracking • Multimodal integration • Quality assurance/quality control

Programming & Scripting: SAS • Python (TensorFlow, nibabel, AMICO, NumPy, scikit-learn) • R • Bash/Shell scripting • MATLAB

Machine Learning & Deep Learning: Deployment of pre-trained deep learning models for neuroimaging registration (SynthMorph, VoxelMorph) • GPU-accelerated inference (CUDA/TensorFlow) • Automated method selection and quality-based fallback logic • Integration of ML components into production Bash pipelines

Statistical Methods: GLM • Mixed effects models • Longitudinal analysis • Multiple comparisons correction • Multivariate approaches • Machine learning

Data Management & Reproducibility: BIDS format specification • BIDS scaffold generation and metadata enrichment • Multi-modal DICOM-to-BIDS conversion tooling (dcm2niix integration, configurable pattern mapping, parallel processing, source archival) • Git/GitHub (CI/CD via GitHub Actions) • Containerized workflows (Docker, Singularity/Apptainer) • Checkpoint-based pipeline resumption • Structured logging (text, HTML, JSONL) • Data documentation & SOPs

Computing Environment: High-performance computing (SLURM) • Containerized pipeline deployment • WSL2 (Windows Subsystem for Linux 2) • NVIDIA GPU computing environments • Hardware benchmarking and computational environment validation for neuroimaging workloads

Data Visualization & Reporting: R (ggplot2), Python (matplotlib) for statistical figures • AFNI, FSL, FreeSurfer, SPM for imaging-specific renders and QC visualization • Automated per-subject QC reports (multi-page PDF with image mosaics, metrics tables, and connectome matrix) • Publication and conference figures

MRI Operations: Siemens 3T Prisma operation (Level 1 certified, 2023) • syngo platform • Protocol implementation • Sequence selection • Human subjects scanning

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PROFESSIONAL EXPERIENCE

Subject Matter Expert, AI Training | 2024–Present
Outlier AI

- Achieved "Oracle" contributor status recognizing sustained high-quality performance across 20+ codenamed research projects and 750+ contribution hours, including multi-phase evaluation workstreams (factuality arbitration, side-by-side quality evaluation).
- Design frontier-level adversarial prompts targeting model failure in neuroscience, biology, clinical medicine, pharmacology, and broader STEM domains; validate failure modes and generate evaluation rubrics specifying reward criteria, accuracy standards, and penalty conditions to guide correct model responses.
- Test agentic AI systems for domain knowledge failures and reasoning errors across multi-step tasks, assessing accuracy and logical coherence in neuroscience and biomedical contexts.
- Review AI model outputs and contributor-generated content for scientific accuracy and alignment with quality standards, providing task-level feedback to quality managers.
- Generate and validate detailed image descriptions and annotations for multimodal AI training initiatives.

Neurotechnology Consultant | 2025
MetaBrain Labs Inc.
Early-stage, pre-funding neurotechnology startup developing an AI-powered cognitive performance platform for the elite athletic performance market

- Drafted a randomized controlled pilot study protocol, including study design, outcome measures and human subjects considerations, for a study planned pending funding.
- Researched IRB options in the US and Europe (institutional partnerships and independent IRBs) for a planned pilot study and shared the findings with the team.

Independent Neuroimaging Software Development | 2024–Present
Open-Source Tools | GitHub: https://github.com/TravisBeckwith

Integrated DTI Processing Pipeline (dwiforge) *(Bash/Python, v2)*
End-to-end diffusion MRI processing and structural connectomics pipeline for BIDS-formatted datasets

- Built automated BIDS-layout validation and intake QC, computing SNR and detecting acquisition parameters to flag malformed inputs before processing begins.
- Developed multi-stage preprocessing incorporating MP-PCA denoising and Gibbs unringing (MRtrix3), N4 bias correction and SynthStrip skull-stripping, and DESIGNER2-based eddy current correction and Rician noise removal, with distortion correction via Synb0-DisCo synthetic-b0 generation and FSL topup field estimation, run in a containerized environment with automatic runtime detection (Docker/Singularity/Apptainer).
- Integrated FreeSurfer recon-all cortical reconstruction with ML-enhanced T1w-to-DWI registration, automatically selecting between SynthMorph (FreeSurfer), VoxelMorph, and ANTs, including GPU-accelerated inference via TensorFlow and quality-based fallback logic.

- Incorporated NODDI microstructure estimation (NDI, ODI, ISOVF) via AMICO, with automated API version handling for cross-version compatibility.
- Implemented SS3T-CSD whole-brain tractography (10M streamlines) with SIFT2 biological filtering, atlas-space parcellation registration, and Desikan-Killiany connectome construction, applying a centered-log-ratio (CLR) transform to correct for the compositional nature of SIFT2-weighted connectivity data in group-level statistics.
- Designed a two-level checkpoint system enabling safe pipeline resumption after interruption, with stage-level tracking and fine-grained progress monitoring.
- Engineered automated per-subject QC reporting generating a multi-page PDF with axial slice mosaics (b0, FA, MD, NDI, ODI), pipeline stage status, white matter metrics table, and structural connectome matrix.

SMILD *(Python)*
Voxelwise reliability framework for diffusion MRI's Standard Model

- Developed a proof-of-concept of SMILD (Standard Model Inter-branch Likelihood Distance), a per-voxel measure of parameter identifiability addressing the branch degeneracy in diffusion MRI's Standard Model; derived estimation theory against primary literature and verified to behave as predicted on synthetic ground-truth data across three falsifiable predictions (Spearman r = −1.00 against branch separation and SNR), with 18/18 unit tests passing on the core package.
- Planned production estimator (RotInv/LEMONADE-based) for real multi-shell dMRI data and validation against the ABCD Study cohort (pending data access). The current release is a citable, open-source package (Zenodo DOI).

BIDS Conversion Tool (BIDS_convert) *(Bash)*
Configurable DICOM-to-BIDS conversion tool for multi-modal neuroimaging data

- Developed a robust DICOM-to-BIDS conversion tool supporting eleven modalities (anat, func, dwi, fmap, perf, PET, MEG, EEG, iEEG, micr, motion) with automatic folder detection, session-aware directory structure handling, and automatic run-numbering for duplicate acquisitions.
- Implemented a configurable pattern-matching system for mapping DICOM folder naming conventions to BIDS modalities, supporting user-editable configuration files with glob-based matching.
- Built parallel conversion capability with configurable job count, source data archival, and safe deletion with post-conversion output verification.
- Automated BIDS metadata enrichment, including JSON sidecar injection (TaskName, RepetitionTime) and full BIDS scaffold generation (dataset_description.json, participants.tsv, README, CHANGES, .bidsignore), with automatic detection and warning of fieldmap sidecars requiring manual IntendedFor assignment.
- Incorporated dry-run mode for full action preview without data modification, and preserved dcm2niix conversion logs for post-hoc QC.

NeuroRig *(Python)*
Hardware diagnostic tool for neuroimaging processing environments

- Built a lightweight diagnostic utility assessing hardware readiness for MRI processing pipelines (FreeSurfer, fMRIPrep, FSL, AFNI), profiling CPU, RAM, disk I/O, and NVIDIA GPU/CUDA availability.

- Implemented WSL2-specific detection that identifies WSL2 environments, reports RAM actually visible to the subsystem, and flags the common case where WSL2's default host-RAM allocation is bottlenecking the user, with guidance for resolving it via .wslconfig.
- Hardened the tool against common failure modes, including GPU query timeouts, disk benchmark failures, and unexpected runtime errors, with graceful degradation and clear diagnostic messaging rather than a crash.
- Built a 20-test suite covering all core functions, with GitHub Actions CI validating against Python 3.9–3.12 on every push and pull request.
- Provided tiered, interpretive output (RAM/CPU/drive capability ratings) mapped to specific neuroimaging processing capabilities.

gistcv (JavaScript)
Resume-to-job-search-term generator

- Built a full-stack tool that parses uploaded resumes (PDF, DOCX, plain text) and uses an LLM to generate tailored job search terms: primary and adjacent job titles, skill keywords, and platform-ready Boolean/X-ray search strings for LinkedIn and Google.
- Implemented a provider-agnostic LLM integration layer supporting interchangeable backends (Google Gemini, Groq, Anthropic) via configuration, with a free-tier default requiring no paid API access.
- Designed dual deployment paths: a Node/Express server for local or hosted use, and a zero-install static web version for non-technical users to run entirely in-browser with their own API key.

fsglean *(Python)*
BIDS-aware extraction of FreeSurfer stats files into analysis-ready tables

- Built a CLI tool replacing manual per-lab parsing scripts for FreeSurfer output, with BIDS-aware traversal handling both nested and flat subject/session layouts across five stats file types (aparc, aparc.a2009s, aparc.DKTatlas, aseg, wmparc) and three cortical atlases (Desikan-Killiany, Destrieux, DKT).
- Generated both long-format output (ready for mixed-effects modeling in R/lme4 or Python/statsmodels) and wide-format output for cross-sectional analyses, with an auto-generated data dictionary and a subject-session manifest flagging missing files.
- Implemented automated QC flagging of outlier sessions (>4 SD from cohort mean per metric) and compatibility with Infant FreeSurfer and dHCP pipeline outputs.
- Covered by a 74-test suite spanning parsing, core logic, I/O, and CLI behavior.

krige-bold (Geostatistical Functional Denoising) *(Python)*
fMRI denoising method combining random-matrix-theory PCA with geostatistical kriging
Status: reference/pilot implementation, not yet validated on real fMRI data

- Developed a denoising method treating BOLD time series as functional curves rather than scalar summaries: Marchenko-Pastur random-matrix theory objectively separates thermal noise from signal-carrying temporal components, and per-component geostatistical kriging spatially reconstructs each retained component in place of a fixed isotropic smoothing kernel.
- Combined two previously uncombined methodologies (random-matrix-theory PCA, established in MRI denoising; functional-PCA-then-kriging, established in environmental geostatistics but not previously applied to neuroimaging) into a single pipeline.

- Verified on synthetic data via a 7-part automated CI test suite, including an ablation demonstrating kriging measurably improves on rank-truncation alone, and a targeted test reproducing a known failure mode (over-retention under spatially correlated noise) alongside two implemented mitigations.
- Documented theoretical grounding, a critical self-analysis of the method's failure modes, and a validation plan for real fMRI data in an accompanying methods document.

Survey ML Toolkit (surveykit-ml) *(Python)*
End-to-end library for survey/questionnaire data combining classical statistics with supervised machine learning

- Built a full survey-analysis pipeline covering data loading and cleaning, exploratory data analysis, statistical testing, machine learning, and automated HTML report generation, exposed via both a Python API and a CLI.
- Implemented psychometric and inferential statistics including Cronbach's alpha, factor analysis, chi-square and proportion tests, and group comparisons with descriptives.
- Built model comparison and training across Logistic Regression, Random Forest, Gradient Boosting, and XGBoost, with stratified k-fold cross-validation, grid-search hyperparameter tuning, and SHAP-based feature importance for model explainability.
- Implemented k-means respondent segmentation with automated optimal-cluster-count detection via silhouette score and the elbow method.
- Covered by a 217-test suite and a three-stage CI/CD pipeline (lint, matrix testing, coverage, and type-checking; automated documentation builds; automated releases).

Documentation & Dissemination

- Published all tools on GitHub with supporting documentation and assigned DOIs; maintains academic portfolio at travisbeckwith.github.io/CV.
- Debug and harden tools through systematic testing (5–217 test cases per project) and, where implemented, continuous integration (NeuroRig, krige-bold, surveykit-ml), resolving edge cases and failure modes to produce dependable, repeatable behavior rather than one-off scripts.
- Additional tools include QSI-extract, a CI-validated package (63 tests) for collating diffusion MRI scalars (NODDI, DTI, tractometry) from QSIPrep/QSIRecon into longitudinal tables; NeuroMediate, a mediation analysis toolkit supporting ROI, voxelwise, and along-tract inference with bootstrap confidence intervals; and NERVES, a microstructural pipeline computing Normalized Diffusion Entropy maps from T1 and DWI data.

Postdoctoral Research Fellow | 2023–2024 (one-year appointment)
Clinical Neurotrauma Lab | Department of Kinesiology | Indiana University, Bloomington, IN

- Completed fMRI preprocessing and initial statistical modeling for a pilot study of ADHD diagnosis as a moderator of cognitive outcomes after repetitive subconcussive head impacts. N-back behavioral data arrived near the end of the appointment, so covariate-adjusted models are pending; manuscript in preparation.
- Independently selected analysis software and preprocessing strategy for this new research context, then developed and executed a task-based fMRI pipeline (fMRIPrep preprocessing, AFNI statistical modeling) for an N-back working memory paradigm in a longitudinal sports injury cohort, including motion parameter review and artifact flagging to ensure data integrity prior to group-level analysis.

- Began independent development of a diffusion MRI processing pipeline extending beyond the lab's existing methods, later formalized as the open-source dwiforge project, alongside custom Bash scripts for DICOM-to-BIDS conversion that became the basis for the BIDS_convert tool.
- Conducted a targeted review of the literature on blood biomarkers of nervous system damage in neurotrauma to understand standard measures and lab collected bloodwork.
- Acquired neuroimaging data as a certified Level 1 MRI scanner operator on a Siemens 3T Prisma (syngo), including protocol implementation, sequence selection, and quality assurance for human subjects scanning.

Career Pause | 2021–2023
Professional hiatus to address family and personal priorities during the COVID-19 pandemic.

Postdoctoral Research Fellow | 2018–2021
Molecular Epidemiology in Children's Environmental Health Training Program | Division of Epidemiology | Department of Environmental Health & Public Health Services | University of Cincinnati College of Medicine, Cincinnati, OH

- Served as principal neuroimaging analyst for three concurrent NIH-funded longitudinal cohorts (Cincinnati Lead Study, CCAAPS, HOME Study), developing and implementing multimodal analysis pipelines for structural MRI, fMRI, DTI, and MRS data.
- Independently initiated a cortical thickness analysis approach for an air pollution cohort, a novel methodological contribution that resulted in a first-author publication and recognition in the CCHMC annual research report.
- Designed and standardized quality control procedures for multimodal neuroimaging data across three concurrent cohorts with varying acquisition parameters, including systematic visual inspection, outlier flagging, and subject-level documentation.
- Provided methodological consultation across research teams, informing study design and acquisition protocol decisions, and translating pipeline outputs into interpretable results for principal investigators.
- Prepared de-identified neuroimaging datasets for external data sharing in support of collaborative meta-analysis research.
- Authored and co-authored six peer-reviewed publications translating pipeline analyses into scientific findings on environmental impacts on brain development.
- Maintained compliance with IRB protocols as listed key personnel, including annual conflict-of-interest assessments and protocol renewals, across three concurrent studies.

Postdoctoral Research Fellow | 2015–2017
Imaging Research Center | Department of Radiology | Cincinnati Children's Hospital Medical Center, Cincinnati, OH

- Served as primary methodological resource for software and pipeline selection, advising principal investigators and research teams on tool suitability for specific study designs and data characteristics.
- Evaluated and compared neuroimaging analysis tools and preprocessing approaches (FSL vs. Tortoise for diffusion imaging, ANTs for registration, smoothing kernel optimization, atlas selection, motion correction thresholds) to inform methodological decisions and optimize pipeline performance.

- Identified and corrected a critical preprocessing error in fMRI analysis scripts acquired from an external laboratory, catching the issue prior to statistical analysis and notifying the originating lab.
- Authored technical documentation for the MRI_Analysis_Calculator ImageJ plugin, creating a standardized processing manual adopted for ongoing lab use.
- Developed a processing protocol for FDG-PET/CT data examining brain activation in a preclinical stress model, contributing to a Society for Neuroscience conference presentation.
- Validated processing consistency across software versions to ensure reproducibility and reliability of longitudinal analysis outputs.

Graduate Research Assistant | 2008–2015
Neuroscience Graduate Program | University of Cincinnati College of Medicine, Cincinnati, OH

- Led dissertation subprojects within the Cincinnati Lead Study examining associations between childhood lead exposure, psychopathic traits, and criminal behavior using voxel-based morphometry and diffusion tensor imaging.
- Developed and implemented a novel VBM methodology for rodent MRI data, establishing standardized templates and automated batch processing procedures adopted for ongoing lab use and contributing to Lindquist et al. (2015).
- Co-authored five peer-reviewed research publications, two review articles, and one book chapter.

Research Assistant & Lab Technician | 2006–2008
Biology of Affect & Motivation Laboratory | Department of Psychology | Bowling Green State University, Bowling Green, OH

- Contributed to 3 publications, 3 conference proceedings, and 7 conference abstracts on emotional processing and affective neuroscience research.
- Executed behavioral experiments using rodent models, collected and scored data, and maintained testing equipment in compliance with IACUC protocols.
- Managed rodent colony including daily husbandry and care.

Undergraduate Research Assistant | 2006–2007
Transition to Parenthood Study | Department of Psychology | Bowling Green State University, Bowling Green, OH

- Transcribed and entered questionnaire data into a research database for a longitudinal study of relationship transitions.

GRANTS & FUNDING

- **NIEHS T-32 Training Grant Fellowship (T32-ES10957)** | 2018–2020
  Molecular Epidemiology in Children's Environmental Health (MECEH) Training Program | University of Cincinnati College of Medicine

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HONORS & AWARDS

Research & Fellowship Awards:

- 2008–2015: Neuroscience Graduate Program Fellowship, University of Cincinnati College of Medicine

Research Recognition:

- 2026: Research cited in federal amicus brief, *American Water Works Association v. EPA*, U.S. Court of Appeals for the D.C. Circuit (see Media Coverage)
- 2020: Featured Research, Department of Radiology, Cincinnati Children's Research Foundation Annual Report: "Reduced Cortical Thickness Associated with Traffic-Related Air Pollution in a Longitudinally Studied Pediatric Cohort"
- 2020: Most-Shared Findings of January 2020, Cincinnati Children's Hospital Medical Center (CCHMC) Research Horizons Blog

Academic Honors:

- 2007: Outstanding Research Award (Year-long Project), Psychology Undergraduate Research Symposium, Bowling Green State University
- Psi Chi International Honor Society in Psychology
- Sigma Tau Delta International Honor Society in English

Scholarships:

- 2001–2002: President's Achievement Scholarship, Bowling Green State University

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RESEARCH AND SCHOLARLY ACTIVITIES

Publications

Peer-reviewed Journals

1. **Beckwith, T.J.**, Dietrich, K.N., Wright, J.P., Altaye, M. and Cecil, K.M. (2021). Criminal arrests associated with reduced regional brain volumes in an adult population with documented childhood lead exposure. *Environmental Research*, 201, p.111559. PMID: 34181918 DOI: 10.1016/j.envres.2021.111559
2. **Beckwith, T.**, Cecil, K.M., Altaye, M., Severs, R., Wolf, C., Percy, Z., Maloney, T., Yolton, K., LeMasters, G., Ryan, P.H. (2020). Reduced Cortical Thickness Associated with Traffic-Related Air Pollution in a Longitudinally Studied Pediatric Cohort. *PLoS One* 15 (1), e0228092. PMID: 31978108 PMCID: PMC6980590 DOI: 10.1371/journal.pone.0228092
   Altmetric Attention Score: 431, placing it in the top 5% of all research outputs and the 99th percentile among outputs of the same age (Altmetric); Dimensions Field Citation Ratio: 11, cited 70 times. Metrics as of September 2026.
3. Fleck, D.E., Eliassen, J.C., Guerdjikova, A.I., Mori, N., Williams, S., Blom, T.J., **Beckwith, T.**, Tallman, M.J., Adler, C.M., DelBello, M.P., Strakowski, S.M., McElroy, S.L. (2019). Effect of Lisdexamfetamine on Emotional Network Brain Dysfunction in Binge Eating Disorder. *Psychiatry Research: Neuroimaging* 286, 53-59. PMID: 30903953 DOI: 10.1016/j.pscychresns.2019.03.003
4. Brunst, K.J., Ryan, P.H., Altaye, M., Yolton, K., Maloney, T., **Beckwith, T.**, LeMasters, G., Cecil, K.M. (2019). Myo-inositol mediates the effects of traffic-related air pollution on generalized anxiety symptoms at age 12 years. *Environmental Research* 175, 71-78. PMID: 31103795 PMCID: PMC6571158 DOI: 10.1016/j.envres.2019.05.009

Altmetric Attention Score: 328 (top 5% of all research outputs; 99th percentile among outputs of same age). Cited 47 times (Dimensions). Metrics as of September 2026.

5. Raghubar, K.P., Lamba, M., Cecil, K., Yeates, K.O., Mahone, E.M., Limke, C., Grosshans, D., **Beckwith, T.J.**, Ris, M.D. (2018). Dose-volume Metrics and their Relation to Memory Performance in Pediatric Brain Tumor Patients: A Preliminary Study. *Pediatric Blood & Cancer* 65, e27245. PMID: 29856521 PMCID: PMC7388179 DOI: 10.1002/pbc.27245

6. **Beckwith, T.J.**, Dietrich, K.N., Wright, J.P., Altaye, M., Cecil, K.M. (2018). Reduced Regional Volumes Associated with Total Psychopathy Scores in an Adult Population with Childhood Lead Exposure. *NeuroToxicology* 67, 1-26. PMID: 29634994 PMCID: PMC6054826 DOI: 10.1016/j.neuro.2018.04.004

7. Lindquist, D.M., **Beckwith, T.**, Cecil, K.M., Sánchez-Martín, F.J., Landero Figueroa, J., Puga, A. (2015). Prenatal and Early Postnatal Lead Exposure in Mice: Neuroimaging Findings. *Quantitative Imaging in Medicine and Surgery* 5 (4), 511-518. PMID: 26435914 PMCID: PMC4559981 DOI: 10.3978/j.issn.2223-4292.2015.07.01

8. Webber, E.S., Mankin, D., McGraw, J., **Beckwith, T.J.**, Cromwell, H.C. (2013). Ultrasonic Vocalizations, Predictability and Sensorimotor Gating in the Rat. *Behavioural Brain Research* 253, 32-41. PMID: 23850353 PMCID: PMC3759254 DOI: 10.1016/j.bbr.2013.07.013

9. Gitelman, D., Klein-Gitelman, M.S., Ying, J., Sagcal-Gironella, A.C., Zelko, F., Beebe, D., **Beckwith, T.**, Parrish, T., DiFrancesco, M., Brunner, H.I. (2013). Brain Morphometric Changes Associated with Child-Onset Systemic Lupus Erythematosus and Neurocognitive Deficit. *Arthritis & Rheumatism* 65 (8), 2190-2200. PMID: 23666759 PMCID: PMC3840703 DOI: 10.1002/art.38009

10. Webber, E.S., Harmon, K.M., **Beckwith, T.J.**, Peña, S., Burgdorf, J., Panksepp, J., Cromwell, H.C. (2012). Selective Breeding for 50 kHz Ultrasonic Vocalization Emission Produces Alterations in the Ontogeny and Regulation of Rough-and-Tumble Play. *Behavioural Brain Research* 229 (1), 138-44. PMID: 22266925 DOI: 10.1016/j.bbr.2012.01.012

11. Harmon, K.M., Greenwald, M., **Beckwith, T.J.**, McFarland, A.M., Cromwell, H.C. (2009). The Effects of Prenatal Stress on Motivation in the Rat Pup. *Stress* 12 (3), 250-8. PMID: 18951246 PMCID: PMC4137965 DOI: 10.1080/10253890802367265

Book Chapters

1. Cecil, K.M., **Beckwith, T.J.** (2013). A Toxicological Model of Frontal Lobe Injury: Childhood Lead Exposure. In A.E. Cavanna (Ed.), *Frontal Lobe: Anatomy, Functions and Injuries*. Hauppauge, NY: Nova Science Publishers, Inc., 57-78.

Invited Papers/Reviews

1. **Beckwith, T.**, Cecil, K.M. (2013). Acquired White Matter Injury: Infection and Inflammatory Processes. *Journal of Pediatric Neuroradiology* 2 (1), 97-108. DOI: 10.3233/PNR-13050

2. Cecil, K.M., **Beckwith, T.** (2013). Advanced Magnetic Resonance Techniques for Evaluating White Matter. *Journal of Pediatric Neuroradiology* 2 (1), 3-15. DOI: 10.3233/PNR-13043

Published Proceedings

1. **Beckwith, T.**, Dietrich, K., Wright, J., Cecil, K. (2020). Adult measures of criminality correlated with reduced regional brain volumes in a cohort with childhood lead exposure. *Neurotoxicology and Teratology*, 79, 106885. PMID: 32298771 PMCID: 7252070 DOI: 10.1016/j.ntt.2020.106885

2. Brunst, K.J., Ryan, P.H., Altaye, M., Yolton, K., Maloney, T., **Beckwith, T.**, LeMasters, G. and Cecil, K.M. (2018). Role of Traffic-Related Air Pollution on Brain Metabolism and Generalized Anxiety in Adolescents. *Annals of Epidemiology*, 28 (9), 660. DOI: 10.1016/j.annepidem.2018.06.059

3. Klein-Gitelman, M.S., Cedeno, A., Baker, A., Zelko, F., Beebe, D., Dina, B., Sagcal-Gironella, A.C.P., **Beckwith, T.**, Brunner, H.I., DiFrancesco, M., Gitelman, D. (2011). Brain Morphometric Changes Associated with Childhood-Onset Systemic Lupus Erythematosus and Neurocognitive Deficit. *Arthritis & Rheumatism*, 63 (S10), S293. DOI: 10.1002/art.33310

4. McFarland, A.M., **Beckwith, T.J.**, Tran, T., Greenwald, M. and Cromwell, H.C. (2008). The Effects of Natural Fluctuations in Early Maternal Care on the Affective and Behavioral States of the Offspring of Long-Evans Rats. *Ohio Journal of Science*, 108 (1), A-19.

5. **Beckwith, T.J.**, McFarland, A.M., Greenwald, M., Harmon, K., Meserve, L.A., Cromwell, H.C. (2007). Effects of Perinatal Perturbations on Social Communication in the Rat Model. *Ohio Journal of Science*, 107 (1), A-17.

Manuscripts In Preparation

1. **Beckwith, T.**, Nowak, M., Kawata, K. Role of ADHD Diagnosis on Working Memory Before and After Repetitive Subconcussive Head Impacts: A Pilot Study

2. **Beckwith, T**. An Integrated Pipeline for Diffusion MRI Processing, ML-Enhanced Registration, Structural Connectomics, and NODDI Microstructure Modeling *(working title; target: JOSS, Frontiers in Neuroinformatics, or NeuroImage)*

3. **Beckwith, T.** Standard Model Inter-branch Likelihood Distance (SMILD): a per-voxel measure of parameter identifiability in diffusion imaging *(working title; target: JOSS, Frontiers in Neuroinformatics, or NeuroImage)*

PRESENTATIONS

Talks & Seminars

1. "Childhood Lead Exposure: Influences on Adult Brain and Behavior." Epidemiology Seminar Series, Department of Environmental Health, University of Cincinnati College of Medicine, 2020

2. "Childhood Lead Exposure: Influences on the Brain and Behavior in Adulthood." Molecular Epidemiology in Children's Environmental Health Training Program, University of Cincinnati College of Medicine, 2019

3. "Brain Differences Associated with Traffic-Related Air Pollution in a Longitudinally Studied Pediatric Cohort." Molecular Epidemiology in Children's Environmental Health Training Program, University of Cincinnati College of Medicine, 2018

4. "A Magnetic Resonance Imaging Study of the Developmental Consequences of Childhood Lead Exposure in Adulthood." Public Dissertation Defense, Neuroscience Graduate Program, University of Cincinnati College of Medicine, 2015

5. "Adult Neuroimaging Abnormalities Associated with Childhood Lead Exposure." Neuroscience Seminar Series, Neuroscience Graduate Program, University of Cincinnati College of Medicine, 2012

6. "White Matter Abnormalities Associated with Childhood Lead Exposure." Neuroscience Seminar Series, Neuroscience Graduate Program, University of Cincinnati College of Medicine, 2012
7. "Childhood Lead Exposure: How it Alters Brain and Behavior in Adults." Neuroscience Seminar Series, Neuroscience Graduate Program, University of Cincinnati College of Medicine, 2011
8. "The Effects of PCB's on Ultrasonic Vocalizations." Neural and Cognitive Sciences Brown Bag Seminar, Department of Psychology, Bowling Green State University, 2007

Conference Presentations

1. Cecil, K.M., **Beckwith, T.J.**, Brunst, K.J., Yolton, K., LeMasters, G., Ryan, P. (2020). Traffic-Related Air Pollution Associated with Altered Neuroimaging Outcomes in a Longitudinally Studied, Pediatric Cohort. Electronic poster presented at the 58th Annual Meeting of the American Society of Neuroradiology, Las Vegas, NV, May 30–June 3.
2. Brunst, K.J., Cecil, K., Altaye, M., **Beckwith, T.**, Yolton, K., LeMasters, G., Ryan, P. (2018). Brain Metabolite Levels May Mediate Traffic-Related Air Pollution Associated Generalized Anxiety Symptoms: Findings from the Cincinnati Childhood Allergy and Air Pollution Study. Poster presented at ISES-ISEE Joint Annual Meeting, Ottawa, Ontario, Canada, August 26–30. O01.04.11
3. Fleck, D., Eliassen, J., Guerdjikova, A.I., Williams, S., Blom, T., **Beckwith, T.**, Tallman, M., McElroy, S. (2018). Effect of Lisdexamfetamine on Prefrontal Brain Dysfunction in Binge Eating Disorder. Poster presented at the Annual Meeting of the American Society of Clinical Psychopharmacology, Miami Beach, FL, May 29–June 1. T7
4. Cecil, K., Maloney, T., Altaye, M., Severs, R., Wolfe, C., Percy, Z., **Beckwith, T.**, Yolton, K., LeMasters, G., Ryan, P. (2017). Traffic-Related Air Pollution Associated with Activation in a Functional MRI Verb Generation Task of a Longitudinally Studied, Pediatric Cohort. Electronic poster presented at the 25th Annual Meeting of ISMRM, Honolulu, HI, April 22–27. 4600
5. Cecil, K., **Beckwith, T.**, Altaye, M., Severs, R., Wolfe, C., Percy, Z., Maloney, T., Yolton, K., LeMasters, G., Ryan, P. (2017). Traffic-Related Air Pollution Associated with Reduced Cortical Thickness and Altered White Matter Organization in a Longitudinally Studied, Pediatric Cohort. Oral Presentation (#0185) at the 25th Annual Meeting of ISMRM, Honolulu, HI, April 22–27. 3615
6. Cecil, K., **Beckwith, T.**, Wright, J., Dietrich, K. (2016). Volumetric and Diffusion Tensor Imaging Outcomes Associated with Adult Measures of Criminality in a Cohort with Childhood Lead Exposure. Presentation at the 54th Annual Meeting of the American Society of Neuroradiology, Washington, DC, May 23–26. O-240
7. Cecil, K., Hugentobler, J., Quatman-Yates, C., Gubanich, P., Altaye, M., Wade, S., **Beckwith, T.**, Kurowski, B. (2016). Spectroscopic Outcomes Associated with an Aerobic Intervention Following Mild Traumatic Brain Injury in Adolescents. Electronic poster presented at the 54th Annual Meeting of the American Society of Neuroradiology, Washington, DC, May 23–26. eP-188
8. **Beckwith, T.**, Cecil, K.M., Wright, J.P., Dietrich, K.M. (2016). Adult Measures of Psychopathy in a Cohort with Childhood Lead Exposure: Volumetric and Diffusion Tensor Imaging Outcomes. Electronic poster presented at the 54th Annual Meeting of the American Society of Neuroradiology, Washington, DC, May 23–26. eP-73
9. Egan, A., Eliassen, J., Norris, M., Lemen, L., LaSance, K., **Beckwith, T.**, Lindquist, D., Ulrich-Lai, Y. (2015). Use of FDG Positron Emission Tomography to Visualize Brain Activation in a Model of Stress Relief by

"Comfort" Food. Poster presented at 45th Annual Meeting of the Society for Neuroscience, Chicago, IL, October 17–21. W25

10. **Beckwith, T.**, Dietrich, K., Cecil, K. (2012). Reduced White Matter Volume Associated with Childhood Lead Exposure. 50th Annual Meeting of the American Society of Neuroradiology, New York, NY, April 21–26. O-506
11. **Beckwith, T.**, Cecil, K., Dietrich, K. (2012). Sex Differences in Diffusion Tensor Imaging Metrics in Adults with Childhood Lead Exposure. 50th Annual Meeting of the American Society of Neuroradiology, New York, NY, April 21–26. O-507
12. Webber, E.S., **Beckwith, T.J.**, Peña, S.R., Cromwell, H.C. (2009). Selective Breeding for Differential Emissions of 50 kHz Ultrasonic Vocalizations: An Examination of Social Recognition and Fear Conditioning. Poster presented at 39th Annual Meeting of the Society for Neuroscience, Chicago, IL, October 17–21. 191.16/EE79
13. Webber, E.S., **Beckwith, T.J.** (2008). Prepulse Inhibition in Rats Selectively Bred for Differential Ultrasonic Vocalization Emission. Poster presented at 38th Annual Meeting of the Society for Neuroscience, Washington, DC, November 15–19. 883.11/TT87
14. McFarland, A.M., **Beckwith, T.J.**, Greenwald, M., Cromwell, H.C. (2008). Effect of Variation in Maternal Care on Early Social Motivation in Rat Pups. Poster presented at 38th Annual Meeting of the Society for Neuroscience, Washington, DC, November 15–19. 795.18/UU36
15. McFarland, A.M., Greenwald, M., **Beckwith, T.J.**, Cromwell, H.C. (2008). The Effects of Natural Fluctuations in Early Maternal Care on Social Motivation in Young Rats. Poster presented at the 20th Annual Convention of the American Psychological Society, Chicago, IL, May 22–25
16. Harmon, K.H., Greenwald, M., McFarland, A.M., **Beckwith, T.J.**, Cromwell, H.C. (2007). The Effects of Mild Prenatal Stress on Offspring Social Motivation. Poster presented at the 37th Annual Meeting of the Society for Neuroscience, San Diego, CA, November 3–7. 644.12/III27
17. McFarland, A.M., **Beckwith, T.J.**, Jolous-Jamshidi, B., Cromwell, H.C., and Meserve, L.A. (2007). Effect of Polychlorinated Biphenyls on Ultrasonic Vocalizations. Poster presented at Annual Bowling Green State University Research Conference, Bowling Green, OH. Also presented at Bowling Green State University Psychology Undergraduate Research Symposium, Bowling Green, OH
18. Harmon, K.H., McFarland, A.M., Greenwald, M., **Beckwith, T.J.**, Cromwell, H.C. (2007). Mild Prenatal Stress Exposure Disrupts Rat Pup Preference for a Maternally Associated Odor. Poster presented at 14th Annual University of Toledo/Bowling Green State University Symposium on Research in Psychiatry, Psychology & Behavioral Science, Toledo, OH. Also presented at the 5th Annual University of Toledo Neuroscience Research Day, Toledo, OH

TEACHING & TRAINING

Teaching Assistant | 2011–2013
Excellence in Science and Learning Program (ExSEL) | University of Cincinnati College of Medicine & Howard Hughes Medical Institute

- Developed and delivered original STEM curriculum over multiple years, including presentations, laboratory exercises, and interactive activities for cohorts of 20 academically selected high school students.

- Mentored and supervised 4 graduate teaching assistants on classroom management and teaching strategies.
- Managed scheduling and logistics for research faculty presentations.

Teaching Assistant | 2012
Fundamentals of Neuroscience I | University of Cincinnati College of Medicine

- Served as a teaching assistant under faculty mentorship as part of the Preparing Future Faculty certificate program, gaining formal training in course design and pedagogy.
- Led discussion sections, laboratory activities, and study sessions for first-year neuroscience graduate students.
- Collaborated with course directors in the development and delivery of neuroanatomy laboratory exercises.

Springboard Coach | 2008
Springboard Program | Bowling Green State University, Bowling Green, OH

- Provided one-on-one mentorship to first-year students as part of a structured peer mentoring program, supporting their transition to university life through guidance on academic advising, time management, and stress management.
- Facilitated group activities and delivered presentations on effective study habits and available campus resources.

MEDIA COVERAGE

Selected media, legal, and policy coverage of Beckwith et al. (2021), "Criminal Arrests Associated with Reduced Regional Brain Volumes in an Adult Population with Documented Childhood Lead Exposure," *Environmental Research*:

Federal Litigation

Brief of American Academy of Pediatrics, et al. as Amici Curiae in Support of Respondents. *American Water Works Association v. United States Environmental Protection Agency*, No. 24-1376, U.S. Court of Appeals for the D.C. Circuit. Filed April 17, 2026. https://www.nrdc.org/sites/default/files/2026-04/2026-04-17_AAP_Corrected_Final_Amicus_Brief.pdf

Legal & Policy Citations

Cykosky, Phoebe. "'We Plead Lead': How Lead Painted Juvenile Crime Rates in the 1990's." *Vermont Law Review Blog*, May 2, 2025. https://lawreview.vermontlaw.edu/we-plead-lead-how-lead-painted-juvenile-crime-rates-in-the-1990s/

Institutional Coverage

"Long-Term Study Documents Link Between Adult Crime and Brain Damage from Childhood Lead Exposure." *Research Horizons*, Cincinnati Children's Hospital Medical Center, October 25, 2021. https://scienceblog.cincinnatichildrens.org/long-term-study-documents-link-between-adult-crime-and-brain-damage-from-childhood-lead-exposure/

Selected international and national media coverage of Beckwith et al. (2020), "Reduced Cortical Thickness Associated with Traffic-Related Air Pollution in a Longitudinally Studied Pediatric Cohort," *PLoS One*:

1. "Heavy traffic pollution may affect kids' brain development." Linda Carroll, *Reuters*, January 28, 2020. https://www.reuters.com/article/us-health-pollution-brain-development-idUSKBN1ZR2KG
2. "High air pollution exposure in 1-year-olds linked to structural brain changes at age 12." *Neuroscience News*, January 26, 2020. https://neurosciencenews.com/air-pollution-brain-changes-15565/
3. "Babies who live near heavily polluted roads have 'less grey matter in their brains' by the age of 12, study finds." Harry Howard, *Daily Mail*, January 25, 2020. https://www.dailymail.co.uk/health/article-7924721/Babies-live-near-heavily-polluted-roads-developed-brains-age-12-study-finds.html
4. "Pollution from traffic fumes can impair children's sight by shrinking their brains." Tom Bawden, *i*, January 24, 2020. https://inews.co.uk/news/environment/pollution-early-exposure-traffic-fumes-sight-shrinking-brains-389887
5. "MRI shows air pollution effect on kids' brain development." Wayne Forrest, *AuntMinnie*, January 24, 2020. https://www.auntminnie.com/clinical-news/mri/article/15625115/mri-shows-air-pollution-effect-on-kids-brain-development
6. "High air pollution exposure in 1-year-olds linked to structural brain changes at age 12." *ScienceDaily*, January 24, 2020. https://www.sciencedaily.com/releases/2020/01/200124155107.htm

Additional coverage appeared in *PsychCentral*, *Deccan Herald*, *The London Economic*, *Air Quality News*, *Medindia*, *Green Car Congress*, *The Hawk*, and *BreezoMeter*.

SERVICE AND LEADERSHIP

Leadership & Committee Involvement

Student Representative | 2012–2013
Neuroscience Graduate Program Admissions Committee | University of Cincinnati College of Medicine

- Proposed and established the student representative position within the NGP by making the case directly to program directors and faculty.
- Evaluated NGP applicants through interviews and assessments, contributing to the student selection process.
- Gathered and presented feedback from current NGP students regarding applicant interactions, providing insights to the admissions committee.

Student Representative | 2012–2013
Network for Neuroscience Discovery Steering Committee | University of Cincinnati College of Medicine

- Invited by NGP directors and faculty to join the Steering Committee following the establishment of the student representative role in the admissions process.
- Presented research to prospective donors, communicating the scientific mission and impact of the program to non-specialist audiences.
- Participated in discussions on the allocation of newly awarded institutional funding, contributing a student perspective to program-level budget decisions.

Graduate Student Recruiter | 2009–2011
Neuroscience Graduate Program | University of Cincinnati College of Medicine

- Served as a point of contact for recruits, fielding questions about the NGP, graduate student life, and living in Cincinnati.
- Facilitated NGP recruitment activities, including escorting candidates to interviews and events.
- Provided feedback on recruitment strategies and candidate assessments to the NGP Admissions Committee.

Grant Review & Judging

Pilot Project Program Reviewer | 2020
Center for Integrative Environmental Health Sciences (CIEHS) | NIEHS P30 Environmental Health Sciences Core Center | University of Louisville

- Reviewed and scored a pilot grant application submitted to a NIEHS-funded Environmental Health Sciences Core Center, evaluating scientific merit and feasibility.

Poster Session Judge | 2019
40th Annual Graduate Student Research Forum | University of Cincinnati College of Medicine

Manuscript Review

Ad Hoc Manuscript Reviewer for:

- *Environment International*
- *NeuroImage*
- *Physiology & Behavior*
- *Psychiatry Research: Neuroimaging*
- *The Lancet Planetary Health*

Professional Affiliations

- American Association for the Advancement of Science (2020–2021)
- Developmental Neurotoxicology Society (2020–2021)
- Indiana Academy of Science (2024–2025)
- National Postdoctoral Association (2018–2021)
- Ohio Miami Valley Society for Neuroscience (2015–2021)
- Society for Epidemiologic Research (2020–2021)
- Society for Neuroscience (2009–2024)